The Seventeen Wallpaper Groups

Tommy Applebaum, Warren Kozak, Nora Wootten

March 16, 2026

Abstract

While wallpaper patterns are a familiar concept, the mathematics behind them are not as commonly understood. Wallpaper groups are a subgroup of the Euclidean group defined on the two-dimensional plane, and are the collection of symmetry groups of two-dimensional repeating patterns. We say that two given wallpapers are "equivalent" if they have the same symmetry. By this definition, we find that there are 17 unique wallpaper equivalence classes (or symmetries) that exist. In the following paper we will define the Euclidean Group, Lattice Groups, and Point Groups, with a later focus on Wallpaper Groups, as well as exploring the classification of the 17 symmetries of wallpaper.

1 Introduction

Identifying repetition and patterns is intrinsic to the human brain. We are programmed to seek coincidence, find overlap in experience, and generally identify when there exists a relationship between elements. In doing so, we are also predisposed to creating patterns.

In our art, tiling, textiles, architecture, and other forms of expression, humans create repeating patterns of shapes. When imposed onto a 2D plane, these patterns can repeat forever. When shifted, rotated, and flipped, these repeating patterns can be laid identically over themselves. When the patterns can be manipulated in this way, we call the repetition of these shapes planar symmetries.

Importantly, these symmetries present in wallpaper. For example, a wallpaper with a checkerboard tiling pattern is composed of squares, which can be shifted along the x or y axis while still preserving the pattern, what we will later define as translational symmetry. Or, a wallpaper patterned with equilateral triangles might preserve rotational symmetry by rotating triangles by 120 degrees around their centers. Finally, in Figure 1, we see three examples of wallpapers that have 90 degree rotational symmetry, which is to say, rotating about the center of the pattern preserves the pattern (this will be covered in more depth, and redefined later).

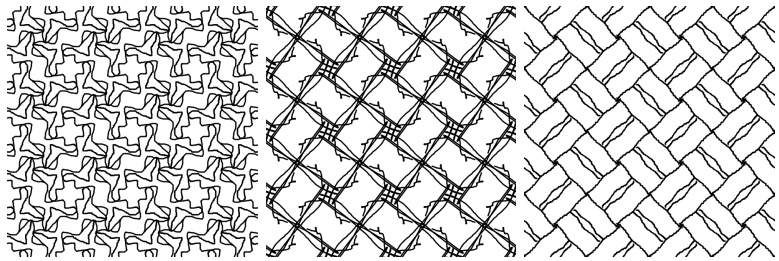
When we begin considering different examples as we are doing now, it becomes clear that there are *many* wallpaper patterns that we can consider. Taking all wallpaper patterns that preserve planar symmetries along the 2D plane, we construct a group that we call the *wallpaper group*. However, as we will later discuss, despite the vast number of repeating wallpaper patterns that exist, there are only 17 planar symmetries.

To begin understanding the wallpaper group and its symmetries, we must first familiarize ourselves with the 2D plane, and as such with the Euclidean Group.

2 The Euclidean Group

2.1 Euclidean Group Properties

The operative word in defining the Euclidean Group is "group." To classify a group we must check that we have a non-empty set containing an identity that is closed under inverses and an associative binary operation. In defining the Euclidean Group, we will start by defining the elements of our non-empty set.



(a) 1 (b) 2 (c) 3

Figure 1: Wallpaper Examples With 90 Degree Rotational Symmetry

Definition 2.1. An *isometry* is a rigid, geometric transformation of a Euclidean space that preserves the Euclidean distance between every pair of points in that space. Note that this transformation is a function $g : \mathbb{R}^2 \to \mathbb{R}^2$.

The Euclidean Group's elements are isometries. We know that this set is non-empty because the set contains the identity transformation, which is the identity element of the Euclidean group. The identity transformation sends every point to itself, leaving the plane unchanged. Depending on notation, we can express it using various names: $f(x) = x$, the identity matrix, etc. To continue our definition of a group, we will examine the binary operation.

Definition 2.2. The *Euclidean Group* is the set of isometries of the plane under composition of functions. The Euclidean Group in the 2 dimensional cartesian plane (the xy plane) can also be expressed as E_2.

We will complete the remaining justification that E_2 is a group in the proof of Claim 2.3, however, first observe that the following notation is used to denote the distance between points x and y: $||x - y||$.

Claim 2.3. *E_2 is a group under function composition.*

Proof. We have already shown that E_2 is a non-empty set and that it contains an identity (the identity transformation). Thus, to show that E_2 is a group under function composition, we still must show that function composition is associative and that E_2 is closed under the binary operation and under inverses.

First, we already know that function composition is associative, which demonstrates that the binary operation defined on E_2 is associative.

Next, we aim to show that E_2 is closed under function composition. Then, consider some functions $g, h \in E_2$ such that g, h are functions from $\mathbb{R}^2 \to \mathbb{R}^2$. The composition of g with h is $g \circ h = g(h(x)) = gh$. Next, consider some points $x, y \in \mathbb{R}^2$. Then, we see that: $||g(h(x)) - g(h(y))|| = ||h(x) - h(y)||$ because g is an isometry, and thus preserves distance between points. Then, also: $||h(x) - h(y)|| = ||x - y||$ because h is an isometry, and also preserves distance between points. Thus, we see that the composition of g with h preserves distance between points, and is itself also an isometry. Thus, $gh \in E_2$, and E_2 is closed under the binary operation.

Next, we aim to show that E_2 is closed under inverses. Then, consider some $g \in E_2$ and its inverse g^{-1}. Then, for some $x, y \in \mathbb{R}^2$, $||g(g^{-1}(x)) - g(g^{-1}(y))|| = ||g^{-1}(x) - g^{-1}(y)||$ because g is an isometry. But, note also that $gg^{-1} = e$, so it also follows that $||g(g^{-1}(x)) - g(g^{-1}(y))|| = ||e(x) - e(y)|| = ||x - y||$. Then, we see that $||g^{-1}(x) - g^{-1}(y)|| = ||x - y||$, and so g^{-1} is also an isometry. As such, $g^{-1} \in E_2$ for all $g \in E_2$, and so E_2 is closed under inverses.

Then, because E_2 is a non-empty set with an identity that is closed under an associative binary operation and inverses, we can conclude that E_2 is a group under function composition, as desired. □

Now that we know that E_2 is a group, we might begin to consider common or familiar elements of that group. We know that rotations, reflections, and translations are all common isometries, because each transformation preserves euclidean distance.

Definition 2.4. A *rotation* is an isometry where an object is rotated around a fixed point contained in its area. For example, the matrix $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ rotates a shape by 90 degrees clockwise.

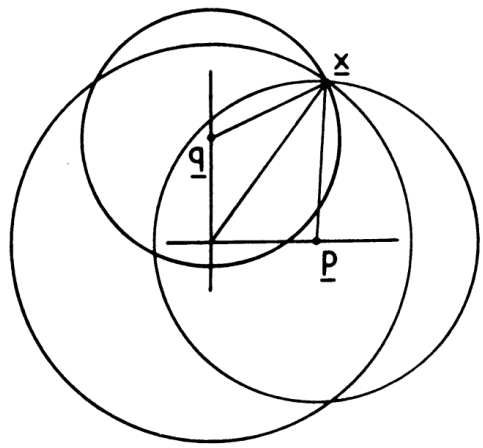


Figure 2: Three Circles With Centers That Are Not Colinear Intersecting at a Single Point

Definition 2.5. A *translation* is an isometry where an object is shifted along the plane in a given direction.

Note that a translation for an entire plane can be defined exclusively based on where it sends the origin. Note that an arbitrary translation is defined as $\tau(x) = x + v$ where v is some translation vector. An example is $\tau(x) = x + 2$, which shifts an object up 2 units.

Definition 2.6. A *reflection* is an isometry where an object is inverted across a line. For example, the matrix $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ reflects across the x-axis because the y component of every point is multiplied by -1.

Claim 2.7 defines what a general element in E_2 is. The proof begins by considering an arbitrary isometry that moves the origin to a given point, v. This isometry achieves this movement through some combination of rotations, reflections, and translations that are undefined because it is arbitrary. Then, we consider a translation τ by v, and observe that this translation would also send the origin to the point v. Then, because τ sends points to v, τ^{-1} sends translated points back to the origin. So, there exists some function $f = \tau^{-1}g$ that fixes the origin. Rearranging the equation, we see that $f\tau = g$. Thus, it suffices to show what kind of isometry f is to show that g is a reflection or rotation followed by a translation. We will do just that in the proof of Claim 2.7.

Claim 2.7. *A general element $g \in E_2$ can be written as either:*

(1) A rotation about the origin and then a translation, or

(2) A reflection about a line through the origin and a translation.

Proof. Consider some arbitrary $g \in E_2$. Define $v = g(0)$. Then, let τ denote a translation by v, which also sends 0 to v. Then, let $f(0) = \tau^{-1}g(0) = 0$. Rearranging the equation, we see that $g(0) = \tau f(0)$. Then, given that τ is a translation, we see that g can be written as a translation and some other element, f. Then, it suffices to show the possible combinations of isometries of f to prove the claim. Thus, we claim that f is either a rotation about 0 or a reflection across a line through 0.

Then, consider $p = (1, 0)$ and $q = (0, 1)$. Every point $x \in \mathbb{R}^2$ is determined by three measurements:

$$||x||, ||x - p||, ||x - q||,$$

where $||x||$ is the distance from the origin, and the other two are the distance between x and p and x and q respectively. These measurements determine every point $x \in \mathbb{R}^2$ because three circles whose centers are not collinear intersect in at most one point, as visualized in Figure 2. That is to say, that any point in $\mathbb{R}^2$ is uniquely determined by its distance from these three points.

Then, because $f(x)$ is an isometry, it will preserve distance between points. So, the position of $f(x)$ can equivalently be determined in reference to it's distance from $f(p)$ and $f(q)$. But, again because $f(x)$ is an isometry,

$$||f(x)|| = ||x||, ||f(x) - f(p)|| = ||x - p||, ||f(x) - f(q)|| = ||x - q||.$$

Thus, once we know $f(p)$ and $f(q)$, we know the impact of f on every point in $\mathbb{R}^2$.

Then, we know that $||f(p)|| = ||p|| = 1$ and $||f(q)|| = ||q|| = 1$ because each are one unit away from the origin on one of the axes. And, $||f(p) - f(q)|| = ||p - q|| = \sqrt{2}$. Thus, the angle between $f(p)$, $f(0)$, and $f(q)$ is a right angle.

Then, let θ be the angle between p, 0, and $f(p)$. To preserve the right angle between $f(p)$ and $f(q)$, either $f(q)$ is a rotation by θ about the origin, or $f(q)$ is a rotation by an angle θ and then a reflection about a line through the origin which subtends an angle of $\frac{\theta}{2}$ with the positive x-axis.

Thus, in either case, we see that f must be a rotation or a reflection. This implies that g is either a reflection or a rotation followed by a translation τ, as desired. □

Thus, every element in E_2 is either a rotation about the origin and then a translation, or, a reflection about a line through the origin and a translation. Then, every element contains some element of translation. However, what if we were to exclusively consider translation elements?

Claim 2.8. *The translations make a up a subgroup T of E_2, that is, $T \leq E_2$.*

Proof. We aim to show that $T \leq E_2$. Thus, we will show that T is non-empty, $T \subseteq E_2$, and use the subgroup criterion to show closure under inverses and the binary operation.

First, we see that the identity translation $\tau(x) = x$ is an element of T. Thus, T is non-empty and contains the identity. Next, consider some $\tau_1, \tau_2 \in T$ such that $\tau_1(x) = x + u$ and $\tau_2(x) = x + v$. These are translations, thus they are isometries, thus they are elements of E_2. Thus, $T \subseteq E_2$.

For τ_1, τ_2, we see that:

$$\begin{aligned} \tau_1\tau_2^{-1}(x) &= \tau_1(x - v) \\ &= (x - v) + u \\ &= x + (u - v). \end{aligned}$$

Then, we see that $\tau_1\tau_2^{-1}$ is of the form of a translation, and is in fact a translation by $u - v$. Then, $\tau_1\tau_2^{-1} \in T$. Thus, by the subgroup criterion, $T \leq E_2$. □

Definition 2.9. An *orthogonal transformation* is a rotation about the origin or a reflection across a line through the origin.

Definition 2.10. The orthogonal subgroup, O, is a subgroup of E_2 that contains all orthogonal transformations.

Then, we see by Claim 2.3 that every element of E_2 can be expressed as a reflection or rotation followed by a translation. Then, it follows that all of E_2 can be expressed by these subgroups, so, we see then that $E_2 = TO$. However, with the exception of the identity translation, any translation moves the origin, while any orthogonal transformation fixes the origin. Thus, the only element in the intersection of these groups is the identity transformation, which implies that elements $g \in E_2$ can only written in one way of the form $g = \tau f$,

Then again consider our arbitrary element $g \in E_2$ such that $g = \tau f$ for some translation τ and some orthogonal transformation f.

Definition 2.11. For $g = \tau f$, if f is a rotation then g is called a *direct isometry.*

Definition 2.12. For $g = \tau f$, if f is a reflection then g is called an *opposite isometry.*

2.2 Euclidean Group as Matrices

By examples given in Definitions 2.4 and 2.6, oftentimes transformation in the plane are best expressed using matrix form. Then, if we continue to consider our arbitrary element $g \in E_2$ where $g = \tau f$, we will now denote a new form of notation with which we can express g.

For $g = \tau f$ where $\tau \in T$, $f \in O$, let $v = \tau(0)$ and let M be the orthogonal matrix that represents the reflection or rotation f in $\mathbb{R}^2$. We can write g using the form:

$$g(x) = v + f_M(x) = v + xM.$$

This formula is an isometry of the plane. Conversely, given some $v \in \mathbb{R}^2$, that is a translational vector in $\mathbb{R}^2$, and some $M \in O_2$, that is a 2×2 transformation matrix, we can define the following:

Definition 2.13. We may now consider each isometry as *ordered pair* $g = (v, M)$ where v is a translational vector and M is a matrix with determinant ± 1. Note that (v, M) is a direct isometry when $\det M = +1$ and an opposite isometry when $\det M = -1$.

Using this definition, we can now redefine each of our previous isometries:

(a) *Translations* by the vector v can be represented by (v, I) for the identity matrix in $\mathbb{R}^2$.

(b) *Rotations* counterclockwise through the origin are $(0, A)$ for some rotation matrix A.

(c) *Reflections* across the line l are $(0, B)$ for some reflection matrix B.

(d) *Rotations* counterclockwise around a specified point c are $(c - f_A(c), A)$ for some rotation matrix A.

(e) *Reflections* across the line m is $(2a, B)$ for $f_B(a) = -a$ where a is a point in the line m.

We will now provide a bit more context about Definitions 2.2 (d) and (e). A rotation about c send each vector x to $c + f_A(x - c)$. Note then that:

$$\begin{aligned} c + f_A(x - c) &= c + f_A(x) - f_A(c) \\ &= c - f_A(c) + f_A(x) \\ &= (c - f_A(c), A). \end{aligned}$$

To understand Part (e) simply, assume that line m is translated onto line l by a translation of $-a$. This is because reflection across the standard line through the origin is easier than translation across an arbitrary line. After translating onto l, perform the reflection on l, and then translate back by a. Thus, we see that:

$$\begin{aligned} a + f_B(x - a) &= a + f_B(x) - f_B(a) \\ &= a - f_B(a) + f_B(x) \\ &= a + a + f_B(a) \\ &= 2a + f_B(x) \\ &= (2a, B). \end{aligned}$$

With this new standardized notation for different isometries, we can introduce a new kind of isometry in E_2.

Definition 2.14. A reflection in a line followed by a translation parallel to the same line is called a *glide reflection*. Let m be the line, then each glide reflection along m has the form $(2a + b, B)$ where $f_B(b) = b$ and $b \neq 0$. Note that a is a point in the line m as in Definition 2.2(e), and b is a translation vector.

Theorem 2.15. *Every direct isometry is a translation or a rotation. Every opposite isometry is a reflection or a glide reflection.*

To prove Theorem 2.15, we consider the two cases mentioned: direct isometries and opposite isometries. For each case, we show that each group only contains two possible geometric transformations. We begin by considering the cases of direct isometries; first we consider the trivial case which ends up being a translation, and then we consider the non-trivial case which we see is a rotation by showing that there exists a fixed point around which the plane rotates.

Then, we will consider the opposite isometries, and examine the possible motions of opposite isometries, which we find are either a reflection or a glide reflection.

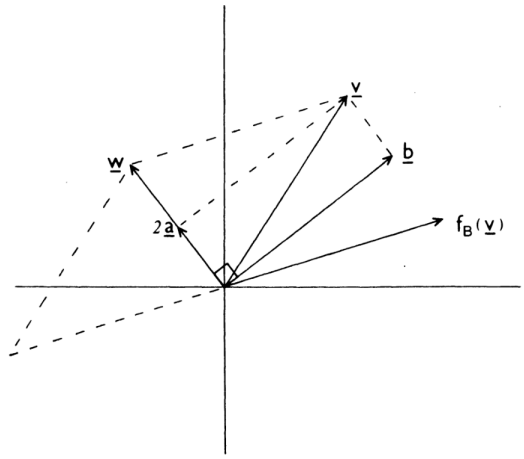


Figure 3: Projecting v onto w

Proof. We will first prove that every direct isometry is a translation or a rotation, and then prove that every opposite isometry is a reflection or a glide reflection.

First, we consider an arbitrary direct isometry, (v, A), where matrix A represents a rotation $0 \leq \theta < 2\pi$. If A represents the rotation $\theta = 0$, then, the only motion is the translation by vector v, and thus, (v, A) is a translation. Then, assume that A represents a rotation $0 < \theta < 2\pi$.

Then, we aim to show that when A represents a rotation $0 < \theta < 2\pi$ that (v, A) is a strict rotation. Thus, we must confirm that there exists a fixed point c about which the plan rotates. We know that c is fixed if it satisfies the equation $c = Ac + v$. Thus, we see that this equation rearranges to $c - Ac = v$ which simplifies to $c(I - A) = v$. Thus, if $c(I - A) = v$ has a unique solution, then these exists one rotation point. The equation only has a unique solution when the determinant of $(I - A)$ is positive.

Then, we will consider the determinant of $I - A$:

$$\det(I - A) = \det \begin{pmatrix} 1 - \cos\theta & \sin\theta \\ -\sin\theta & 1 - \cos\theta \end{pmatrix} = 2 - 2\cos\theta.$$

Note that $-1 < \cos\theta < 1$ because $0 < \theta < 2\pi$. Thus, $2 - 2\cos\theta$ is positive, so $\det(I - A)$ is positive, and there exists a unique c that satisfies the equation $c - f_A(c) = f_{I-A}(c) = v$. Then, the specified direct isometry is the rotation $(c - f_A(c), A)$ about this point c.

Thus, because our given direct isometry was arbitrary and this covers the cases, we can conclude that every direct isometry is a translation or a rotation.

Next, we consider the opposite isometries. Let (v, B) be an arbitrary opposite isometry. First we consider the trivial case where if $f_B(v) = -v$ then we have reflection across the line m for which $a = \frac{v}{2}$. Thus, the opposite isometry is a reflection as desired.

Next, consider when $f_B(v) \neq -v$. Then, the motion is not entirely reduced to a reflection. We set $w = v - f_B(v)$ and observe that:

$$\begin{aligned} f_B(w) &= f_B(v - f_B(v)) \\ &= f_B(v) - f_B^2(v) \\ &= f_B(v) - v = -w. \end{aligned}$$

Then, projecting v onto w, as in Figure 3, gives the projection formula $(\frac{v.w}{||w||^2})w$. Thus, our opposite isometry is a glide reflection given by $(2a + b, B)$, where $2a = (\frac{v.w}{||w||^2})w$, and $b = v - 2a$. Then, our opposite isometry is a glide reflection, as desired.

Thus, because our given opposite isometry was arbitrary and this covers the cases, we can conclude that every opposite isometry is a reflection or a glide reflection. $\square$

But, we can simplify the way that we express isometries even further.

Exercise 2.16 (Exercise 24.4)**.** *Show that every isometry can be expressed as the product of either two or three reflections.*

Proof. First recall that isometries are closed under composition. Thus, the composition of isometries is some other isometry, and thus a composition of isometries can also be defined as one of our four main isometry groups: reflections, rotations, translations, and glide reflections. Then, to show that every isometry can be expressed as the product of either two or three reflections, it is sufficient to consider each isometry in cases.

First, most simply, we consider a reflection. Note that reflecting three times across the same line is equal to reflecting once across that line. As such, a reflection can be expressed using three reflections.

Next, we consider a rotation. Reflecting once about a line, and again across another intersecting line creates a rotation. A reflection across two intersecting lines rotates points by 2θ centered at the intersection of the lines, where θ is the interior angle between the lines. As such, a rotation can be expressed using two reflections.

Third, we consider a translation. A translation can be expressed as two reflections across parallel lines. Note that in this case, a point is moved along a reflectional trajectory, and again across the same trajectory about the second line. Thus, c is sent to $f_2(f_1(c))$ where f_1, f_2 are reflections. But, the net movement is along some vector, which is equivalent to a translation along that vector. As such, a translation can be expressed using two reflections.

Finally, we consider our final isometry, a glide reflection. A glide reflection can be expressed as a combination of a reflection and a translation. Recall that a translation can be expressed as a reflection across two parallel lines. Then, reflecting across a third, perpendicular line, combines our translation and reflection to sufficiently create a glide reflection. As such, a glide reflection can be expressed using three reflections.

Thus, we see that by cases, every isometry type can be expressed using two or three reflections. Thus, every isometry can be expressed using two or three reflections. □

3 Lattices and Point Groups

Now that we have established the Euclidean Group, and defined some important properties and notation, we can move on to a discussion about a subgroup that arises when we define two dimensional repeating patterns on the Euclidean plane.

Definition 3.1. Any two-dimensional repeating pattern is known as a **wallpaper pattern**. Refer to Figure 4 to see an example of such a pattern. The symmetry groups that arise from wallpaper patterns are subgroups of E_2, whose elements are the isometries of the plane that send a given pattern to itself, thus preserving the pattern.

Now, we can use the notation established in the previous section to discuss isometries. We can denote every isometry in the Euclidean plane as (v, M), with $v \in \mathbb{R}^2$ and $M \in O_2$, where O_2 is the orthogonal group in two dimensions. This can be interpreted as the set of all rotations and reflections that fix the origin. Note that the notation $g = (v, M)$ represents the transformation of some point $x \in \mathbb{R}^2$ by $g(x) = Mx + v$. If we have another transformation $g_1(x) = M_1x + v_1$, then we have

$$g(g_1(x)) = M(M_1x + v_1) + v = M(M_1x) + M(v_1) + v = MM_1x + (v + Mv_1).$$

So, the matrix part of the transformation is now MM_1, and the vector part is $v + Mv_1$. Using our notation, this can be represented as $(v + f_M(v_1), MM_1)$. Now that we have our formula for isometry composition, we can also define the function $\pi : E_2 \to O_2$ by $\pi(v, M) = M$. We can prove this is a homomorphism, because for $g = (v, M)$ and $g_1 = (v_1, M_1)$, we have

$$\begin{aligned}\pi(gg_1) &= \pi(v + f_M(v_1), MM_1)\\ &= MM_1\\ &= \pi(g)\pi(g_2)\end{aligned}$$

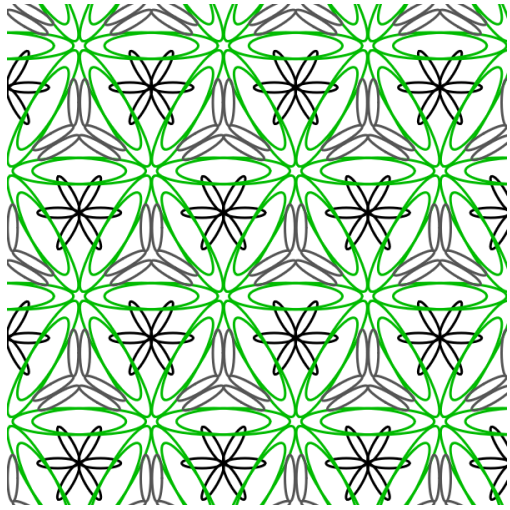


Figure 4: An example of a wallpaper pattern.

Now that we have defined this function π, we have a way to isolate the orthogonal part of any isometry, and strip away any translation information. This allows us to analyze the underlying symmetry of a pattern, without worrying about the pattern's position in Euclidean space.

Definition 3.2. If G is a subgroup of E_2, we can say that $H = G \cap T$, that is that H is the subgroup of all isometric translations in G. We can also say that $J = \pi(G)$, so J is the subgroup of all orthogonal translations in G. We call H the **translation subgroup** of G, and J the **point group** of G.

Translation subgroups and point groups of a wallpaper group G will become important later in this section when we attempt to describe the properties of subgroups of the Euclidean plane, and isometries between subgroups. Now that we have defined the translation subgroup of G and the point group of G, we can construct the definition of a wallpaper group.

Definition 3.3. We say that a subgroup of E_2 is a **wallpaper group** if its translation subgroup can be generated by two distinct translations, and it has a finite point group.

Now that we have defined a wallpaper group, we apply this definition and consider an example that is generated by two glide reflections. To show that it is in fact a wallpaper group, we will need to show two things. First, that the translation subgroup contains two linearly independent vectors, and that the point group is finite. We will compose the glide reflections to attempt to get two linearly independent translations, and we will analyze the number of elements in point group of the subgroup.

Example 3.4. The subgroup of E_2 generated by the glides $g(x, y) = (-x, y+1)$ and $h(x, y) = (-x+2, y+1)$ forms a wallpaper group (From Exercise 25.1 (i)).

We can begin by determining whether the translation subgroup contains two distinct translations. Since this subgroup is generated by two glides, we can compose the glides to find translations. Consider $g \circ h$ and $h \circ g$. We have

$$g \circ h = g(h(x, y)) = g(-x + 2, y + 1) = (x - 2, y + 2), \text{ and}$$
$$h \circ g = h(g(x, y)) = h(-x, y + 1) = (x + 2, y + 2).$$

So the translation vector of $g \circ h$ is $t_1 = (-2, 2)$, and the translation vector of $h \circ g$ is $t_2 = (2, 2)$. These are linearly independent, as they are not scalar multiples of each other (they are in fact perpendicular). Thus, the translation subgroup contains two distinct translations.

Now, we must show that the point group is finite. We can see that the linear part of g and h is simply

$$M = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}.$$

This corresponds to a reflection across the y-axis. Now, composing this linear transformation with itself, we get

$$M^2 = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I_2$$

Thus, for $H = G \cap T$ we have $H = \{M, I_2\}$. Since the translation subgroup contains two distinct translations, and the point group is finite with order 2, the subgroup of E_2 generated by g and h forms a wallpaper group.

Example 3.5. The subgroup of E_2 generated by the translation $\tau(x, y) = (x + 1, y)$ and a reflection in the y-axis does not form a wallpaper group (From Exercise 25.1 (ii)).

We are given one horizontal transformation of $(1, 0)$, and a reflection $r(x, y) = (-x, y)$. We need to test if we can create a linearly independent transformation by composing the generators. Conjugating, we can see that

$$(r \circ \tau \circ r^{-1}) = (-(-x + 1), y) = (x - 1, y).$$

This gives us a translation vector of $(-1, 0)$, which we can see is a scalar multiple of the original translation vector. Thus we cannot create a second linearly independent vector, so this is not a wallpaper group. Geometrically, this makes sense, as a vector with only a x component, when reflected across the y-axis, will never be able to gain a y component.

Example 3.6. The subgroup of E_2 generated by reflection in the x-axis and anticlockwise rotation through $2\pi/3$ about the point $(0, 1)$ forms a wallpaper group (From Exercise 25.1 (iii)).

Let us define R to be the rotation about $(0, 1)$ through $2\pi/3$, S to be the reflection in the x-axis, and $R_{mirror} = S \circ R \circ S^{-1}$. Conjugating R with S gives us a rotation by $-2\pi/3$ about the point $(0, -1)$. We can create the first translation $\tau_{v_1} = R_{mirror} \circ R$. Since the exact opposite rotation occurs about a different point, the net effect is no rotation and instead a translation by the vector v_1. Then, we can say that $\tau_{v_2} = R \circ T_{v_1} \circ R^{-1}$. Geometrically, this rotates the translation vector v_1 by $2\pi/3$ creating a new, linearly independent translation vector v_2, and thus we have two linearly independent translations, τ_{v_1} and τ_{v_2}. Additionally, the point group is generated by a reflection and a single $120°$ rotation, which forms the finite dihedral group D_6. Thus, this subgroup forms a wallpaper group.

Example 3.7. The subgroup of E_2 generated by reflection in the x-axis, reflection in the y-axis, and reflection in the line $x + y = 1$ forms a wallpaper group (From Exercise 25.1 (iv).

If we can construct two parallel lines of reflection, then we can get a translation. Reflecting the line $y = 0$, or the x-axis, across the line $x + y = 1$, we get the line $x = 1$. Now we have two parallel vertical reflections across $x = 1$ and $x = 0$. Composing these reflections, we get a translation τ_1 by $(2, 0)$. Similarly, we can construct two horizontal lines of reflection. Reflecting the line $x = 0$ across the line $x + y = 1$ yields the line $y = 1$. Now we have reflections across $y = 0$ and $y = 1$. Composing these reflections together, we get a translation τ_2 by $(0, 2)$. Thus we have two linearly independent translations. Additionally, the point group is generated by reflections intersecting at $45°$ and $90°$. Similar to the last example, these form the finite dihedral group D_8, which is the symmetries of a square.

Notice that the order of the point group in Example 3.7, and it is made up of only the identity isometry and a reflection in one direction. This will become important later on as we attempt to classify wallpaper groups.

Now, let G denote a wallpaper group with a translation subgroup H and a point group J, and let L be the orbit of the origin under the action H on $\mathbb{R}^2$. We know that L contains two linearly independent vectors, as the translation subgroup H is generated by two independent vectors. Thus we know L spans all of $\mathbb{R}^2$. Now select two vectors, $a, b \in L$, such that a of minimum length in L and b is as small as possible, and a and b are skew to each other (not parallel nor intersecting).

Theorem 3.8. *The set L is the lattice spanned by a and b. That is to say, L consists of all linear combinations $ma + nb$ where $m, n \in \mathbb{Z}$.*

To prove Theorem 3.8, we will use the strategy of proof by contradiction. If we assume some arbitrary element that is in L but not in the lattice, then we can analyze the size of the edge between a corner of the lattice and x, and then derive a contradiction. The key to this proof is our assumption that a is of minimum length in L, and that b is skew to a with a size as small as possible. These facts allow us to derive a contradiction about the size of this new vector we have created.

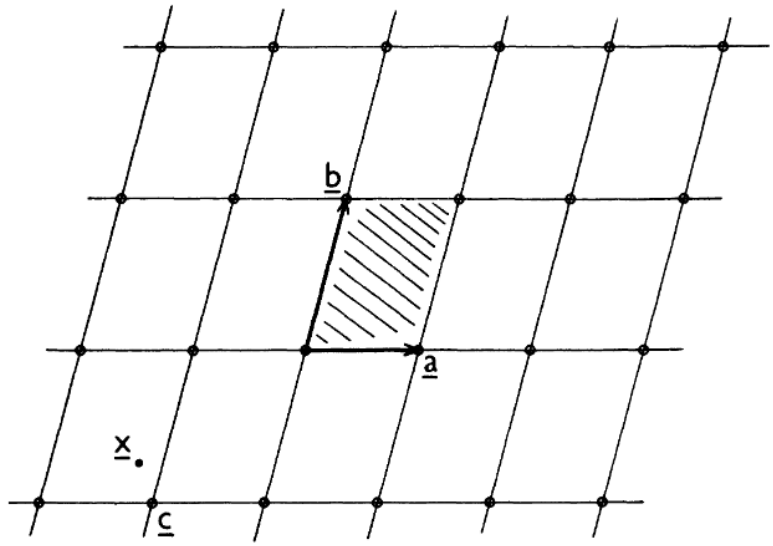


Figure 5: Lattice spanned by a and b with off-lattice point x

Proof. Consider the map $\varphi : T \to \mathbb{R}^2$ by $(v, I) \mapsto v$. We know this is an isomorphism, meaning that L is a subgroup of $\mathbb{R}^2$. Then L is closed under addition, and since multiplication is just repeated addition, we know that $ma + nb \in L$. So the entire lattice spanned by the vectors a and b is in L. Now we must show these are the only points in L. Assume, for contradiction, that there is some $x \in L$ such that x is not on the lattice. Refer to Figure 5 to see an example of vectors a and b with a point x. Then we can select a parallelogram from the lattice where x is inside, such that x is as close to some vertex c of the lattice as possible. Now consider the vector $x - c$. We know that $x - c$ is not the 0 vector, and we know that $||x - c|| \neq b$ and $||x - c|| \neq a$. We also know that $||x - c|| < b$. Since $x \in L$ and $c \in L$, we know $||x - c|| \in L$, so we cannot have $||x - c|| < a$ since we assume a was of minimum length. So then we have $||a|| \leq ||x - c|| < ||b||$. We would have $x - c$ is skew to a, however this is also a contradiction since we assumed b was skew to a and as small as possible. Thus no point $x \in L$ can be off the lattice, so L is spanned by a and b. □

In the proof of Theorem 3.8, we defined a parallelogram with edges consisting of the skew vectors a and b. The shape of this parallelogram determines different properties of the generated lattice, and we can classify these different lattice forms into five categories based on the relative lengths of a and b, and the lengths of their sum and difference.

Definition 3.9. The five different unique lattices are defined as follows:

1. Oblique: $||a|| < ||b|| < ||a - b|| < ||a + b||$
2. Rectangular: $||a|| < ||b|| < ||a - b|| = ||a + b||$
3. Centered Rectangular: $||a|| < ||b|| = ||a - b|| < ||a + b||$
4. Square: $||a|| = ||b|| < ||a - b|| = ||a + b||$
5. Hexagonal: $||a|| = ||b|| = ||a - b|| < ||a + b||$

Exercise 3.10 (Exercise: 25.13)**.** *Let G be a wallpaper group which has a square lattice. What possibilities are there for the point group of G?*

Proof. Imagine a single square within the square lattice with horizontal and vertical vectors a. Define the origin such that it lies at the center of the square. Then, we can imagine the point group as the group of symmetries which maintain the single lattice cell. First, we can see that we can rotate the cell about the origin by the angles $0, \pi/2, \pi, 3\pi/2$. Similarly, we can reflect the square lattice cell about lines with angles

$0, \pi/4, \pi/2, \pi$. We can see that this is the same as the dihedral group D_8 represented in matrix notation with determinant equal to 1. $\square$

Theorem 3.11. *The point group J acts on the Lattice L.*

To prove this, we just simply show that the lattice L is closed under the action of J, that is that for $x \in L$ and $M \in J$, we have that $f_M(x) \in L$ as well. To do this, we can utilize a group theory trick, by conjugating the isometry with translation component x with the isometry with orthogonal component M, noticing that H, or the translation subgroup is normal inside of G. Then the desired result follows.

Proof. We know that $J \leq O_2$, or the orthogonal group in two dimensions, so J acts on the plane in the same way that O_2 does. To show J acts on L, a subgroup of $\mathbb{R}^2$, we must show for some $M \in J$ and $x \in L$, that $f_M(x) \in L$. Let $\tau = (x, I)$, or the pure translation by the vector x, and for some $g = (v, M) \in G$, we have $\pi(g) = M$. We know that H is the kernel of the homomorphism π defined earlier in the chapter, so we can now observe that H is normal in G. Then by the definition of a normal subgroup, we know $g\tau g^{-1} \in H$. Expanding this out, we get

$$\begin{aligned} g\tau g^{-1} &= (v, M)(x, I)(-f_M^{-1}(v), M^{-1}) \\ &= (v, M)(x - f_M^{-1}(v), M^{-1}) \\ &= (v + f_M(x - f_M^{-1}(v)), MM^{-1}) \\ &= (v + f_M(x) - v, I) \\ &= (f_M(x), I) \end{aligned}$$

So, we know that $(f_M(x), I) \in H$, and thus $f_M(x)$ is a point in the lattice L. So J acts on the lattice L. $\square$

Now that we have establish this important theorem, we can begin to draw restrictions on the possible rotations in the point groups of wallpaper groups. We can use Theorem 3.11 to have rotations in J act on an arbitrary vector of minimal length in L, and show that only rotations of certain amounts respect this minimality. Rotation amounts and the orders of rotations are inherently connected, so this gives us restrictions on the possible orders of rotations in the point group J.

Theorem 3.12. *The order of a rotation in a wallpaper group can only be 2, 3, 4, or 6.*

To prove this theorem, we can begin by defining what a general power of some rotation with order q is. Then, we assume some vector of minimum length in the lattice of the wallpaper group G, and show how any rotation with order equal to 5 or greater than 6 causes a contradiction by creating a vector in the lattice that is smaller than a.

Proof. Since the point group of any wallpaper group G is finite, we know that every rotation in G has finite order. Let us say some rotation has an order of q. Then, we can say that some power of this rotation is counter clockwise rotation of $\frac{2\pi}{q}$. From the formula for a counter clockwise rotation matrix in two dimensions, we get

$$M = \begin{pmatrix} \cos\frac{2\pi}{q} & -\sin\frac{2\pi}{q} \\ \sin\frac{2\pi}{q} & \cos\frac{2\pi}{q} \end{pmatrix}.$$

As we did in the proof of Theorem 3.8, let a be a non-zero vector of minimum length in the lattice L. From Theorem 3.11, we know that J acts on L, so $f_M(a) \in L$. Assume, for contradiction, that q is greater than 6. Then $\frac{2\pi}{q} < 60°$, and thus $f_M(a) - a \in L$ but is smaller than a, which is a contradiction. Now assume that q is 5. If we take $f_M^2(a) + a$, we once again are left with a vector that is less than a, which is again a contradiction. Thus the only possible orders are 2, 3, 4, or 6. $\square$

Corollary 3.13. *The point group of a wallpaper group is generated by a rotation through one of the angles $0, \pi, 2\pi/3, \pi/2, \pi/3$ and possibly a reflection*

Proof. This statement follows directly from Theorem 3.12. We have that the orders of a rotation in a wallpaper group can only be 2, 3, 4, or 6. Thus, we have that the only possible rotations are of the angles, once simplified, are π, $\frac{2\pi}{3}$, $\frac{\pi}{2}$, and $\frac{2\pi}{3}$. A rotation by 0 serves as the identity of the point group. $\square$

Now that we have built up some of the facts surrounding wallpaper groups, we must prove a central theorem that will allow us to thoroughly classify wallpaper groups. By showing that certain wallpaper groups are not isomorphic to each other, we can show that there are exactly 17 kinds. But first, we must establish that an isomorphism between wallpaper groups respects the structure of the groups.

Theorem 3.14. *An isomorphism between wall paper groups takes translations to translations, rotations to rotations, reflections to reflections and glide reflections to glide reflections.*

To prove Theorem 3.14, we will use the orders of different kinds of isometries. We will first derive a contradiction that translations cannot map to glide reflections, as this violates the fact that isomorphisms respect commutativity of elements. Then, we will use a similar argument to show that reflections must map to reflections, by deriving a contradiction that a glide reflection cannot map to a half turn, as an element with an infinite order cannot map to an element with order 2 under an isomorphism.

Proof. Suppose we have some isomorphism $\varphi : G \to G_1$ between two wallpaper groups, and suppose we have a translation $\tau \in G$. We know that translations and glide reflections have infinite order, so we know that $\varphi(\tau)$ has to also be a translation or a glide reflection. Suppose, for contradiction, that $\varphi(\tau)$ is a glide reflection. Now, let $\tau_1 \in G_1$ be a translation that does not commute with $\varphi(\tau)$, meaning that τ_1 is not parallel to the line of reflection for $\varphi(\tau)$, and let $g \in G$ such that $\varphi(g) = \tau_1$. Since τ_1 is a translation in G_1, we know that it has infinite order, and thus $g \in G$ also must have infinite order, so it is either a translation or a glide reflection. Applying a translation twice yields another translation, and applying a glide reflection twice also yields a translation, so we know that $g^2 \in G$ is a translation. All translations in wallpaper groups are commutative, so we know that g^2 commutes with τ. Then since G is isomorphic to G_1, we know that $\varphi(g^2) = \tau_1^2$ must commute with $\varphi(\tau)$. This is impossible, however, since we assumed that τ_1 wasn't parallel to the reflection line of $\varphi(\tau)$, so then τ_1^2 isn't either. Thus $\varphi(\tau)$ must be a translation, so translations map to translations and glide reflections map to glide reflections. Now we will show that reflections map to reflections. Let $g \in G$ be a reflection. We know g has order 2, since all reflections have an order of 2. Thus $\varphi(g)$ also has an order of 2. The only possibilities for $\varphi(g)$ are reflections and half-turns, as these are the only elements of wallpaper groups that have an order of 2. Suppose, for contradiction, that $\varphi(g)$ is a half-turn. Now suppose $\tau \in G$ is a translation that is not perpendicular to the reflection line of g. Now consider $\varphi(\tau g) = \varphi(\tau)\varphi(g)$. We know from part 1 that $\varphi(\tau)$ is a translation, and we assumed that $\varphi(g)$ was a half-turn. A translation composed with a half-turn is just another half-turn, so $\varphi(\tau g)$ has an order of two. However, τg is a glide reflection, which has infinite order. This is a contradiction, as an element of G with infinite order cannot map to an element with an order of 2. Thus, under φ, reflections map to reflections. Rotations are the only class of isometry left in G and G_1, so it follows that rotations in G map to rotations in G_1. □

Corollary 3.15. *If two wallpaper groups are isomorphic then their point groups are also isomorphic.*

Proof. Corollary 3.15 follows from the proof of Theorem 3.14. Let G, G_1 be wallpaper groups with translation subgroups H, H_1 and point groups J, J_1. Suppose we have an isomorphism $\varphi : G \to G_1$. By Theorem 3.14, we have that $\varphi(H) = H_1$. Thus, we also know that $G/H \cong G_1/H_1$. We have that $G/H \cong J$ and $G_1/H_1 \cong J$, since the wallpaper group G modded out by all translations is exactly the point group J. Thus, we know that $J \cong J_1$. □

4 Wallpaper Group Notation

In the next section, we will prove that there are 17 wallpaper groups. However, first we need to establish the notation that will be used. Each group will be a combination of the symbols: p, m, g, c, and the integers: 1,2,3,4,6.

Definition 4.1. A *primitive cell* is a parallelogram with vertices at lattice points that contains no lattice points in the interior. A lattice that be formed from repeating primitive cells is called a *primitive lattice*.

We know from section 3 that oblique, rectangular and square lattices are trivially primitive lattices, and hexagonal lattices can be formed from 3 parallelograms where our vectors a, b are defined as the edges of one parallelogram in the lattice. Wallpaper groups formed from these lattices will begin with a p.

Definition 4.2. A *centered cell* is a parallelogram with vertices at lattices that contain a lattice point at the center. A lattice formed from repeated centered cells is called a *centered lattice*.

We will denote wallpaper groups formed from centered lattices will begin with a c. The centered rectangular lattice is the only centered lattice that we can form wallpaper groups from.

The letters m and g refer to the opposite isometries when applied to the lattice maintain structure where m refers to a reflection and g refers to a glide reflection. From 3.12, we know that rotations in wallpaper groups must have order 2, 3, 4, or 6, so we will use those integers to denote when a rotation of that order is the rotation of the highest order in the wallpaper group. Finally, 1 refers to the identity.

5 Types of Wallpaper Groups

For this section, we will always define our lattice by the linear combination of 2 vectors, a and b. Write a such that it lies on the positive x-axis and b such that it lies in the first quadrant. Additionally, we will define general matrix representation for isometries that will be commonly used throughout the classification of each wallpaper group.

Define A_θ to be the matrix representation of a counterclockwise rotation of θ about the origin, and B_φ be the matrix representation of a reflection over a line with angle $\varphi/2$ above the x-axis. We will select a finite point group and then using our translational subgroup generated by the linear combination of a and b, giving us a wallpaper group by 3.3, test to see what new wallpaper groups can be generated.

We will do this by determining what non-translational elements exist in the wallpaper group, and then partitioning the group using the translational subgroup, H, to determine the general elements of each group. We are able to do this as H is a normal subgroup, the kernel of $\pi : E_2 \to O_2$ from section 3. We will then in Section 6 use our understanding of these general elements to show each of our generated wallpaper groups are not isomorphic to any other.

5.1 Case 1: Oblique Lattices

We can see that for an oblique lattice, L, the only isometries which maintain the cell in the lattice will be the identity isometry and a rotation of π about the center of the cell. Then, we know the point group of any wallpaper group with an oblique lattice must be a subgroup of $G = \{\pm I\}$.

*p*1. This is the simplest of all wallpaper groups. The point group of G is $J = \{I\}$, and the translational subgroup of G is generated by a and b. We can see any element of G can be written in the form $(ma+nb, I)$ for $n, m \in \mathbb{Z}$. □

*p*2. Now, look at the point group $J = \{\pm I\}$. We know that H is normal, so we can partition our set G, by H to give us H and $H(0, -I)$ where $(0, -I)$ is a rotation of π about the origin. Then, the general element of G is $(ma + nb, I)$ or $(ma + nb, -I)$ for $m, n \in \mathbb{Z}$ which gives us a wallpaper group. □

From now on, for all wallpaper groups with just rotations, we will only denote the point group and the cosets as the only difference between them will be the number of cosets.

5.2 Case 2: Rectangular Lattices

When our lattice, L, is rectangular, we now have 4 isometries which maintain L: the identity, a half turn, a reflection in the x-axis and a reflection in the y-axis. That gives us a point group $J \leq \{I, -I, B_0, B_\pi\}$. Since we want unique wallpaper groups, we will ignore *p*1, *p*2 as we have already shown they are wallpaper groups. We will first look at subgroups of J with the point group $\{I, B_0\}$

pm. Consider $J = \{I, B_0\}$ and G contains a reflection over the horizontal axis. Then, our reflection has the form $(0, B_0)$ and the translational subgroup is still generated by a and b so a general non-translation in G is written in the form $(ma + nb, B_0)$ for $m, n \in \mathbb{Z}$. We can see when $m = 0$, we have a reflection and for all other $m \in \mathbb{Z}$ there will be a glide reflection in G where ma will be the translation portion of the glide. □

pg. Continuing with $J = \{I, B_0\}$, but now G does not contain a reflection over the horizontal axis. Then, it must be true that G contains a glide reflection. We know that a sequence of 2 glide reflections is equivalent to a translation so we can write our glide reflection in the form $(1/2ka, B_0)$ for $k \in \mathbb{Z}$. Note that a is the specified vector as a glide reflection will translate parallel to the axis of reflection. We will now show that k must be odd via contradiction.

Assume k is even. We know that $(-1/2ka, I)$ is a translation in G so we can get

$$(-1/2ka, I)(1/2ka, B_0) = (-1/2ka + f_M(1/2ka), IB_0) = (0, B_0) \tag{1}$$

which is a reflection contradicting our original assumption, so k must be odd as then $(-1/2ka, I)$ is not a translation in G. Then, we can see $(-1/2(k-1)a, I)(1/2ka, B_0) = (1/2a, B_0)$ which means a general non-translation element of G is of the form $(ma + nb, I)(1/2a, B_0) = ((m + 1/2)a + nb, B_0)$. □

Look at what happens when we have the point group $J = \{I, B_\pi\}$. This is completely equivalent to rewriting the above cases with our horizontal and vertical axes switched, so the only shift will be that the general element in *pg* will have a glide that affects the vector b in stead of a. This shows $J = \{I, B_0\}$ and $J = \{I, B_\pi\}$ are isomorphic.

We will now assume that our point group is $J = \{I, -I, B_0, B_\pi\}$ for the rest of the wallpaper groups with a rectangular lattice.

p2mm. Let G contain reflections across both the horizontal and vertical axes. Then, G must contain $(0, B_0)$ and $(0, B_\pi)$. The only important note is that $(0, B_0), (0, B_\pi)$ implies the existence of $-I \in G$ as we can generate the half-turn by applying a reflection over the x-axis and then the y-axis to get $(0, B_\pi)(0, B_0) = (0, -I)$. We can then partition the set G using H to get the cosets H, $H(0, B_0)$, $H(0, B_\pi)$, and $H(0, -I)$.

All elements in the first coset are translations. Every element in the second coset is of the form $(ma + nb, B_0)$ for $m, n \in \mathbb{Z}$. However, similar to what we saw in pm, since G has no glide reflections, the general element is (nb, B_0), and the general element of the third coset is (ma, B_π) for $m \in \mathbb{Z}$. Finally, a general element of the fourth coset is simply $(ma + nb, -I)$. □

p2mg. Let G contain reflection across the x-axis, but not the y-axis. Then, we must have a glide reflection in the vertical mirror. Using a similar argument as in *pm* and *pg*, and selecting an origin at the intersection of the reflecting axes, we get that $(0, B_0), (1/2b, B_\pi) \in G$. The product of the reflections gives us $(1/2b, B_\pi)(0, B_0) = (1/2b, -I)$ or the half turn about $1/4b$. We now partition G using the translation subgroup, H to get the cosets H,$H(0, B_0)$, $H(1/2b, B_\pi)$, and $H(1/2b, -I)$.

We know all elements in the first coset are translations. Then, a general element in the second coset is (nb, B_0) for $n \in \mathbb{Z}$ as any vector ma would imply that the element is a glide reflection. A general element of the third coset is $(ma+(n+1/2)b, B_\pi)$ which are all vertical glides that lie on lattice points or midway between them. Finally, a general element of the fourth coset is every half turn about the points $ma + (n + 1/2)b$ for $m, n \in \mathbb{Z}$. □

As with *pm* and *pg*, we do not need to show that there is a wallpaper when the reflection and glide reflection axes are swapped as then the wallpaper groups are still isomorphic.

We will leave the case of *p2gg* to the reader as the method will remain the same as the case of *p2mg* except the cosets will contain only glide reflections and the rotation will be about the point $(m+1/2)a+(n+1/2)b$.

p2gg. Suppose $J = \{I, -I, B_0, B_\pi\}$ and there are no reflections in G. □

5.3 Case 3: Square Lattices

We will now move on to looking at square lattices. Then, the set of isometries that maintain the lattice is the dihedral group D_8, so we know any point group must be a subgroup of this dihedral group. Additionally, the only way to generate new wallpaper groups is to have $A_{\pi/2}$, a rotation by $\pi/2$, be included in the point group.

p4. When $J = \langle A_{\pi/2} \rangle$, there are no reflections in the point group, so our group G has cosets H, $H(0, -I)$, $H(0, A_{\pi/2})$, and $H(0, A_{3\pi/2})$. □

For wallpaper groups with the point group $J = D_8$, we will only look at the case of $p4gm$ as the case $p4mm$ will rely on simply generating the reflection in the group and then composing it with the rotation $(a, A_{\pi/2})$, a quarter turn, to generate every element.

p4mm. When $J = \langle A_{\pi/2}, B_0 \rangle$ where B_0 can be realized using a reflection across the x-axis. □

p4gm. Let $J = \langle A_{\pi/2}, B_0 \rangle$ except we will now assume that there does not exist $(0, B_0) \in G$. Select an origin point such that $(0, A_{\pi/2})$ is in G. Let $(\lambda a + \mu b, B_0)$ be the isometry that corresponds to $B_0 \in J$. If we apply $(\lambda a + \mu b, B_0)$ twice, we get the translation $(2\lambda, I)$ so 2λ must be an integer. We can see that 2λ must be odd as when it is even, we can create the reflection $(\mu b, B_0) = (-\lambda a, I)(\lambda a + \mu b, B_0)$. Now we will use this to generate our general elements of G. First, we have the glide $((1/2a + \mu b, B_0)$ and we can also generate $(0, A_{\pi/2})(1/2a + \mu b, B_0) = (1/2b - \mu a, B_{\pi/2})$. When, we square this glide to get the translation, $((1/2 - \mu)(a + b), I)$, we can see that $1/2 - \mu$ must be an integer which allows us to create the glide $(1/2a + 1/2b, B_0) = ((1/2 - \mu)b, I)(1/2a + \mu b, B_0)$. This gives us the 8 right cosets of H: H, $H(0, A_{\pi/2})$, $H(0, A_\pi)$, $H(0, A_{3\pi/2})$, $H(1/2a + 1/2b, B_0)$, $H(1/2a + 1/2b, B_{\pi/2})$, $H(1/2a + 1/2b, B_\pi)$, and $H(1/2a + 1/2b, B_{3\pi/2})$.

Note that the cosets $H(1/2a + 1/2b, B_0)$, $H(1/2a + 1/2b, B_\pi)$ tells us that this will have glide symmetry along both the horizontal and vertical axes, however, we can also generate reflections in the coset $H(1/2a + 1/2b, B_{\pi/2})$. A general element of this coset is $((m + 1/2)a + (n + 1/2)b, B_{\pi/2})$ for $m, n \in \mathbb{Z}$ which we can rewrite in the form $(1/2(m + n + 1)(a + b) + 1/2(m - n)(a - b), B_{\pi/2})$. Take $m + n + 1 = 0$ and we generate reflections across 45 degree lines that are midway between the lattice points. For nonzero $m + n + 1$, when $m - n$ will be odd, we will have those same axis of reflection just with glides and when $m + n$ is even, we will have glides across the diagonal of the square lattice. $H(1/2a + 1/2b, B_{3\pi/2})$ will give us the reflections and glides along the 135 degree lines. □

Note that since the existence of a singular glide necessarily implies that there exists a reflectional symmetry in $p4gm$, there does not exist a wallpaper group $p4gg$.

5.4 Case 4: Hexagonal Lattices

We can finally move on too looking at the wallpaper groups with a hexagonal lattice. Then, we see that all isometries which maintain the lattice will be the dihedral group D_{12}, and we will look only at point groups which contain rotations of order 3 or order 6 as all other points groups have already been shown to have corresponding wallpaper groups. First we will look at point groups which contain $A_{2\pi/3}$ and not $A_{\pi/3}$.

p3. Let $J = \langle A_{2\pi/3} \rangle$. Then, we have a wallpaper group with only rotations of order 3 and the identity so our cosets of the translational subgroup are of the form H, $H(0, A_{2\pi/3})$, and $H(0, 4\pi/3)$. □

For both $p3m1$ and $p31m$, the point groups will be generated by a rotation of order 3 and a reflection. The difference will be the angle of the line or reflection. We will only look at he case of $p31m$, but the strategy can applied to determine the general elements of the other wallpaper group. The strategy will also be the same for $p6mm$ further below except it must be done in both directions.

p31m. Suppose that $J = \langle A_{2\pi/3}, B_{\pi/3} \rangle$. Then, select the origin such that $(0, A_{2\pi/3}) \in G$. Now, let $(\lambda a + \mu b, B_{\pi/3})$ be a reflection or glide that corresponds to $B_{\pi/3}$ in the point group.

We can see that when we square this isometry we get

$$(\lambda a + \mu b, B_{\pi/3})^2 = ((\lambda + \mu)(a + b), I) \tag{2}$$

which implies that $\lambda + \mu$ must be an integer as the squared reflection must be a translation in G. We will now use a sequence of isometry compositions to show that λ and μ must be integers.

First, composing our reflection and the rotation $(0, A_{2\pi/3})$ gives us $(0, A_{2\pi/3})(\lambda a + \mu b, B_{\pi/3}) = (\lambda(b - a) - \mu a, B_\pi)$ which a reflection over the vertical axis. Then, applying that twice will give us the translation $(\lambda(b - a) - \mu a, B_\pi)^2 = (2\lambda b - \lambda a, I)$ so λ is an integer and by proxy, μ is an integer. That tells us that B_0 must be realized by a reflection as $(0, B_{\pi/3}) = (-\lambda a - \mu b, I)(\lambda a + \mu b, B_{\pi/3})$.

This gives us the cosets $H, H(0, A_{2\pi/3}), H(0, A_{4\pi/3}), H(0, B_{\pi/3}), H(0, B_\pi), H(0, B_{5\pi/3})$. □

p3m1. When $J = \langle A_{2\pi/3}, B_0 \rangle$, we would define the same origin and $(\lambda a + \mu b, B_0) \in G$ and then use function compositions to show that $(0, B_0) \in G$ which would give us 6 cosets, H, $H(0, A_{2\pi/3})$, $H(0, A_{4\pi/3})$, $H(0, B_0)$, $H(0, B_{2\pi/3})$, $H(0, B_{4\pi/3})$. □

Then, we now can look at the subgroups which contain $A_{\pi/3}$.

p6. (*p6*) When $J = \langle A_{\pi/3} \rangle$, then we can have a wallpaper group with only rotations of order 6 and the identity so our cosets of the translational subgroup are H, $H(0, A_{\pi/3})$, $H(0, A_{2\pi/3})$,..., $H(0, A_\pi)$. □

p6mm. Let $J = \langle A_{\pi/3}, B_0 \rangle \cong D_{12}$. The strategy will stay the same as with *p31m* where we will define our origin such that $(0, A_{\pi/3} \in G$. Then, we will define $(\lambda a + \mu b, B_0)$ and $(\lambda' a + \mu' b, B_{\pi/3})$ and then show that by composing reflections and rotations we can see that $(0, B_0), (0, B_{\pi/3}) \in G$. Then, we would partition into 12 right cosets of H where each coset is an isometry in D_{12}. □

Note that with the case of *p31m*, and similarly for *p3m1* and *p6mm*, the existence of the wallpaper group implied that there must be a reflection. This tells us that we do not have any wallpaper groups that contain only glides with no reflections unlike with square and rectangular lattices where this was possible.

5.5 Case 5: Centered Rectangular Lattices

We will now move from using primitive cells to a centered cell. Specifically, the centered rectangular lattice. We can see that the only isometries that will maintain the structure of the lattice will be contained in the same point group as a rectangular lattice, $J = \{I, -I, B_0, B_\pi\}$

cm. Suppose $J = \{I, B_0\}$ and that (v, B_0) is the isometry corresponding to B_0 in the point group. Then, (v, B_0) is either a reflection across a horizontal line or a glide along a horizontal line. We will select a point to be our origin such that $2v$ is a multiple of a, and we know that the vertical direction is determined by $2b - a$ due to the structure of the lattice. Then, $2v = ka$ for $k \in \mathbb{Z}$. We will define the non-translation elements of G using cases.

Case 1: If k is even, then the reflection $(0, B_0) = (1/2ka, I)(-1/2ka, B_0) \in G$ as $(1/2ka, I)$ is a translation in G and $(1/2ka, B_0)$ is a glide reflection in G. Then, we can see the general non-translation element will be $(ma + nb, B_0) = (m + 1/2n)a + 1/2n(2b - a), B_0)$ for $m, n \in \mathbb{Z}$. When n is even, we will have the set of all reflections and glide reflections which axes pass through lattice points. When n is odd, we will have glides which pass midway between lattice points where the translation portion of the glide will be $1/2a$.

Case 2: If k is odd, then we can see $(1/2(2b - a), B_0) = (-1/2(k + 1)a + b, I)(1/2ka, B_0) \in G$. Note that this is also a reflection as $1/2(2b - a)$ is a perpendicular translation to the axis of reflection. □

Similar to with *pm*, wallpaper groups with the point group $J = \{I, B_\pi\}$ will be isomorphic to *cm*.

c2mm. Suppose $J = \{I, -I, B_0, B_\pi\}$. The same method as in *cm* can be applied except in this case it must be done in both the horizontal and vertical axes to show that we once again only get reflections. □

Note that unlike *pm* and *p2mm* there exist glide reflections where the component translation and reflection is not contained within the wallpaper group. This occurs when n is odd and k is even in the case of *cm* as $(1/2a, I)$ is not a translation in G.

6 Proving Wallpaper Groups Are Unique

In this section, we will shift our focus from defining each of the wallpaper groups and in stead show that each wallpaper group is not isomorphic to any other wallpaper group. This section will rely heavily on 3.14, so we will simply mark when we are applying this theorem to avoid repetition. Additionally, we know per the contrapositive of 3.15 that if 2 point groups are not isomorphic, then the wallpaper groups will not be isomorphic, so we can section our proofs of uniqueness to only point groups with the same finite order.

Theorem 6.1. *No two of p2, pm, pg, cm are isomorphic.*

Proof. Per 3.14, any isomorphism between wallpaper groups will send translations to translations, reflections to reflections, rotations to rotations, and glide reflections to glide reflections. That directly shows that only pm and cm could be isomorphic. Imagine we take a glide reflection in cm where the component translation and reflection are not contained within the wallpaper group. Then, since all glides in pm contain their component parts, we can see that cm and pm must not be isomorphic. □

Theorem 6.2. *No two of $p2mm, p2mg, p2gg, c2mm$ are isomorphic*

Proof. This proof will be almost exactly the same as for 6.1 using 3.14 and then decomposing glide reflections to see that $c2mm$ can not be isomorphic to any of the others. □

Theorem 6.3. *$p4mm$ is not isomorphic to $p4gm$.*

Proof. All rotations of order 4 in $p4mm$ can be written as a sequence of 2 reflections while the lack of reflections over 1 axis makes this not necessarily true for all rotations in $p4gm$ which implies $p4mm$ is not isomorphic to $p4gm$. □

Theorem 6.4. *$p3ml$ is not isomorphic to $p31m$.*

Proof. Look at the rotation $(a, A_{2\pi/3})$. We will show that in $p31m$ this rotation can be written as a sequence of 2 reflections, but this is not true in $p3m1$. First we know

$$(a, I)(0, B_\pi)(0, B_{\pi/3}) = (a, B_\pi)(0, B_{\pi/3}) = (a, A_{2\pi/3}) \tag{3}$$

which is a way of writing $(a, A_{2\pi/3})$ as a composition of 2 reflections. However, this is not necessarily true for $p3m1$. Assume for the sake of contradiction that $(a, A_{2\pi/3}) = (v, B_\theta)(v', B_\varphi)$ is a composition of 2 reflections. We know that 2 reflections will give us a rotation angle double the angle between the 2 reflection lines. This means that we can look at this as a sequence of 3 cases: $\theta = 2\pi/3, \varphi = 0$, $\theta = 4\pi/3, \varphi = 2\pi/3$, and $\theta = 0, \varphi = 4\pi/3$.

Let us look at the case $\theta = 2\pi/3, \varphi = 0$. We know that our translation vectors v, v' can be written of the form $v = \lambda(b - 2a)$ and $v' = \mu(2b - a)$ as they must be perpendicular to their axis of reflection as they are not glides. No we can complete

$$(\lambda(b-2a), B_{2\pi/3})(\mu(2b-a), B_0) = (\lambda(b-2a) + f_{B_{2\pi/3}}(\mu(2b-a)), A_{2\pi/3}) = ((\mu-2\lambda)a + (\lambda+\mu)b, A_{2\pi/3}) \tag{4}$$

which is equal to $(a, A_{2\pi/3})$, so we can generate a system of equations to solve for λ, μ. Our system is $\mu - 2\lambda = 1$ and $\lambda + \mu = 0$. When solved, we get $\lambda = -1/3$ and $\mu = 1/3$, but this means that we can generate the translation $(v', B_0)(0, B_0) = (1/3(2b - a), I)$ which is a contradiction as $2/3, -1/3 \notin \mathbb{Z}$. Our other cases, will simply be the same as this case, but rotated $2\pi/3$. Thus, we can not factor the reflection $(a, A_{2\pi/3})$ as a multiple of 2 reflections in $p3m1$. Then, via 3.14, we can see that $p31m$ and $p3m1$ are not isomorphic. □

Thus, we have now shown that the 17 wallpaper groups classified in the last section are each unique groups.

7 Wallpaper Examples With Classification

Finally, in this section, we will present some examples of geometric patterns that can be classified as wallpaper groups, along with their classification within the wallpaper group structure we describe in this paper.

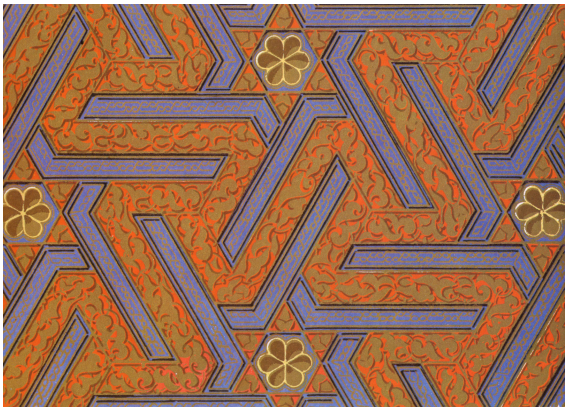


Figure 6: An example of a p6 tiling pattern

Example 7.1. Figure 6 shows wall paneling from the Alhambra palace in Spain [Wik26]. This particular pattern is classified as p6. The p in the name of this wallpaper group indicates that it is formed out of primitive cells, and the 6 indicates that the only symmetry that this pattern has is 6-fold rotational symmetry. If we set the origin of this pattern to be the center of one of the flower petal designs, we can see that the pattern will overlay itself a total of 6 times as it is spun around, thus indicating its 6-fold rotational symmetry. This pattern does not have any reflectional symmetry, nor any glide-reflection symmetry.

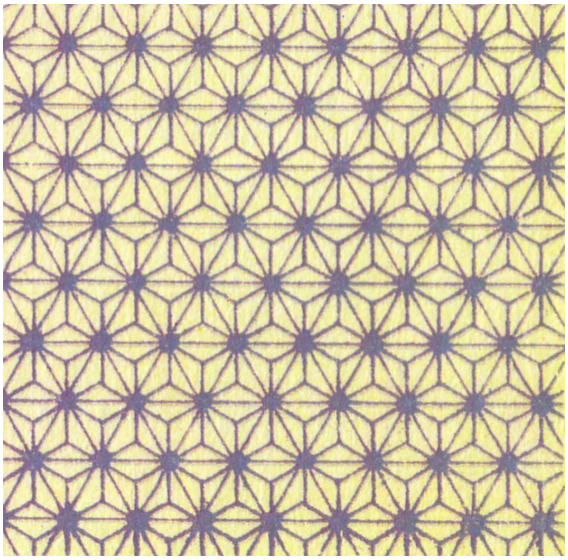


Figure 7: An example of a p6m tiling pattern

Example 7.2. Figure 7 shows a pattern on a piece of painted porcelain from China. This design is classified as a p6m wallpaper wallpaper group. Similar to the pattern shown in Figure 6, this pattern exhibits 6-fold rotational symmetry. However, in addition to the 6-fold rotational symmetry, this pattern also has a vertical axis of symmetry. Reflecting across this vertical axis preserves the pattern, and thus the classification of this wallpaper is of the group p6m.

Figure 8: An example of a p2 tiling pattern

Example 7.3. Despite their name, wallpaper groups do not only appear on art. In Figure 8, we can see an example of a p2 wallpaper group on a wire fence. This indicates that it is made of primitive cells, and only includes 2-fold rotational symmetry. At first glance, this wire fence may look like it has more than just 2-fold rotational symmetry, or at least may have some reflective symmetry. However, we can see that the corners of the interlocking chains are oriented a particular way, meaning that this pattern in fact only has 2-fold rotation symmetry. A rotation by 180° will preserve the pattern, and no other rotation will.

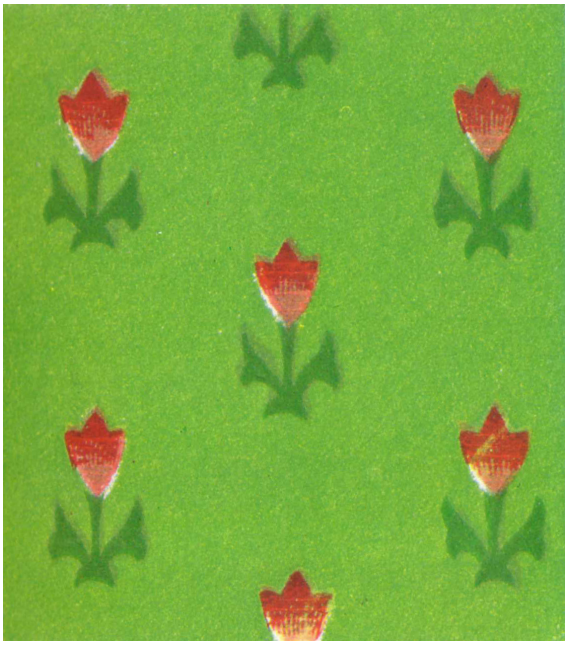


Figure 9: An example of a cm tiling pattern

Example 7.4. Figure 9 shows a Persian tapestry with a repeating flower design. This pattern belongs to the wallpaper group cm. The c in the name indicates that the cells that make up this wallpaper are centered, as one of the translation vectors a or b lies in the center of the cell rather than solely on the borders of the cell. This pattern has a vertical axis of symmetry, and also has glide-reflection symmetry. Reflecting any one of the flowers to the right or left, followed by a translation up or down results in the pattern being preserved.

Notably, solely a reflection about the glide-reflection axis or a translation will not preserve the pattern, and thus are not elements of the wallpaper group cm.

8 Bibliography

References

[Arm88] Mark. A Armstrong. *Groups and Symmetry (Undergraduate Texts in Mathematics)*. Springer, 1988.

[Sas20] Vivek Sasse. *Classification of the 17 Wallpaper Groups*. 2020. URL: `https://math.uchicago.edu/~may/REU2020/REUPapers/Sasse.pdf`.

[Wik24] Wikipedia. *Euclidean group*. 2024. URL: `https://en.wikipedia.org/wiki/Euclidean_group`.

[Wik26] Wikipedia. *Wallpaper group*. 2026. URL: `https://en.wikipedia.org/wiki/Wallpaper_group`.